

Project

The purpose of this project is to let you design an autopilot for a jet fighter's lateral dynamical motion around a straight leveled (horizontal) symmetric trajectory at constant velocity.

The lateral equations of motion are provided by the following state-space model

$$\begin{bmatrix} \dot{\beta} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} -0.575 & 0 & -1 & 0.0536 \\ -300 & -3.03 & 2.0 & 0 \\ 68 & 0.045 & -2.4 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta \\ p \\ r \\ \phi \end{bmatrix} + \begin{bmatrix} -0.078 \\ 64.4 \\ 5.0 \\ 0 \end{bmatrix} \delta_a \quad (1)$$

$$\dot{\delta}_a = 5(\delta_c - \delta_a) \quad (2)$$

where β [rad] is the side-slip angle; p [rad/sec] is the roll rate; r [rad/sec] is the yaw rate; ϕ [rad] is the roll angle. Equation (2) is the dynamics equation for the ailerons deflection δ_a [rad], and δ_c [rad] is the command input to the ailerons servo-mechanism. The aircraft is equipped with a vertical gyro that measures the roll angle ϕ and with a rate gyro that measures the roll rate p . The critical control requirement is that the controlled system shall track a step command of 30 degrees for the roll angle with satisfactory damping and settling time. A non-critical requirement is that the ailerons deflection shall not exceed 5 degrees (up or down).

1. Write the above equations in a five-dimensional state-space representation. Analyze the poles of the system: time constant, damping and natural frequency. The complex conjugate poles are known as the *Dutch Roll mode*, while the slow real pole is the *Spiral mode*, and the other two real poles are related to the *Roll mode* and to the ailerons servo mode. Identify each of these modes on a pole/zero map.
2. Feedback control design using Root-Locus
 - (a) Assuming that your measurement is the roll angle ϕ only and your command input is the ailerons deflection command δ_c , design a Proportional controller such that the roll angle ϕ tracks a step command of 30 degrees. Choose a gain in the controller that produces a satisfactory damping and a natural frequency as large as possible for the dominant poles. Of what type is the closed-loop system? Does this controller improve the properties of the Dutch Roll mode? Analyze the performances of your design by checking at least the closed-loop dynamic response to a 30 degrees step in the desired roll angle, the steady-state tracking error, etc.
 - (b) In order to improve the damping of the system, you suggest using the measurement of the gyro rate p . For your design, you start by first closing the loop on the roll rate p and then on the roll angle ϕ . Check the influence of the roll rate feedback on the Dutch Roll mode and on the Spiral mode. Analyze the performances of your design by checking at least the closed-loop dynamic response and the steady-state tracking error. Compare this design with the Proportional Control design.
 - (c) Assume again that your measurement is the roll angle ϕ only and replace the Proportional-Derivative controller with a Phase Lead Compensation Network in order to improve the damping. Check the trade-off between damping and steady-state tracking error by comparing several designs. Choose one of the designs and compare its performances with those of the Proportional and Proportional-Derivative controllers.

3. Feedback control design in State Space

- (a) Design a full state feedback controller using Pole Placement. Requirements: perfect tracking of a 30 degrees step command in ϕ ; all the closed loop poles shall satisfy: $2 \leq \omega_n \leq 3$ [rad/sec]; one of the poles shall be real and the others shall have damping coefficients $0.5 \leq \zeta \leq 0.7$.
 - (b) Design an observer-based controller with the same requirements in control performances as described in (a) assuming roll angle measurements only.
 - (c) The roll angle measurement device fails but the roll rate gyro is available. If possible, design an observer-based state feedback control with the same requirements as in (a).
 - (d) Design a stationary Linear Quadratic Servo-controller for the lateral motion tracking a 30 degrees step command in ϕ . Check its performances by simulating several dynamic responses with various initial conditions.
4. Provide a short comparison between the various controllers: transient and steady-state performances, model assumptions, physical intuition, computational load, implementation practicality, ...

Good Luck !!!